

Question	Marking Guidance	Mark	Comments
03.1	1. As surface area to volume ratio increases, metabolic rate increases OR (Humans with) a large surface area to volume ratio have a high/fast metabolic rate; 2. (A large(r) surface area to volume ratio will) lose more heat; 3. (A high(er) rate of metabolism/respiration) releases/provides/replaces heat OR (A high(er) rate of metabolism/respiration) maintains body temperature;	3 (3 x AO1)	Accept the converse for all marking points 1. Accept the relationship is proportional 1. Ignore 'directly' if prefixing proportional 1. Accept the relationship is positively correlated 2. Accept lose heat faster/more easily 3. Accept (a higher rate of metabolism/respiration) releases energy OR produces /generates heat 3. Reject produces energy/ heat energy

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03.2	Person A and correct ratio of 23.7:1 OR 23.8:1 = 3 marks;;; Person A and correct ratio (23.75349227:1) not to 3 sf = 2 marks Person A and correct answer not expressed as a ratio = 2 marks Person A and 2.14:0.09 = 2 marks Person B and correct ratio of 23.7 OR 23.8:1 = 2 marks Person B and 26.0:1 = 2 marks (correct calculations, wrong person) Person A and answer not expressed as a ratio (23.75349227) and not to 3 sf = 1 mark Evidence of 2.14 and 1.61 in working = 1 mark Evidence of 0.09 and 0.06(2) in working = 1 mark 1.61:0.06(2) = 1 mark	3 (2 x AO1, 1 x AO3)	Ignore named person alone If no person remove 1 mark

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03.3	Due to a typographical error, this question has been discounted and all candidates will receive 1 mark, even the non-attempts.	1 (1 x AO1)	

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03.4	(Gill) lamella(e) and (gill) filament(s);	1 (1x AO1)	Accept primary and secondary lamellae Ignore (gill) arch and (gill) rakers

